



AI in Finance

Transforming banking, investment, risk
management and services

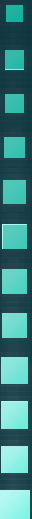
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AI Defined

AI uses algorithms and data to automate pattern recognition and decision-making. In finance, AI encompasses machine learning, deep learning, NLP, and rule-based systems that learn from transaction, market, and customer data to produce actionable outcomes. Example: a fraud-detection model analyzes 12 months of logs to learn normal cardholder behavior; when a large foreign purchase is followed by *r



AI Transforms Finance



In banking, real-time fraud detection and AI-powered chatbots handle routine queries, reducing risk and customer support costs. In lending, alternative-data credit scoring and automated loan processing speed approvals. In markets and risk, algorithmic trading, stress-testing with non-linear factors, and robo-advisors for auto rebalancing.



AI Pipeline

Define
objective

Data
gathering

Feature prep

Model
training

Validation &
deployment

Monitoring &
governance

Set a measurable goal (e.g., reduce fraud false positives by 30%) to guide all decisions.

Collect and label transactions, customer profiles, and external signals to support training.

Engineer time windows and behavioral aggregates to capture relevant patterns.

Choose supervised or unsupervised approach and backtest on historical data.

Run A/B tests; productionize with monitoring; start in alert-only mode.

Track drift, enable human review, and log compliance for ongoing control.

Real-Time, Predictive Wins

Fraud control relies on nightly batch rule checks and manual review, which causes delays and reactive responses. Credit assessment depends on static credit bureau scores and paper underwriting, slowing decisions and limiting data scope. Customer service operates on business hours with human agents, 延

Fraud is scored in real time using behavioral baselines with automated alerts, enabling immediate intervention. Credit decisions leverage alternative data and automated scoring for faster approvals. Customer service runs 24/7 with chatbots that escalate complex cases to humans as needed.

AI transforms workflows from manual, batch, and reactive to automated, real-time, and predictive.

AI Impact Metrics

Illustrative outcomes showing how AI improves speed, accuracy, and cost-efficiency.

60%

Fraud investigation time reduced ~60% through automated triage

30%

False positives reduced ~30% due to improved detection accuracy

40%

Operational costs for routine tasks lowered ~40% via automated processing

Governance Essential

AI in finance introduces data privacy, bias, cybersecurity, and regulatory risks that require governance, transparency, and strong controls. Apply data minimization, encryption, and anonymization to protect client information. Mitigate bias with fair testing and diverse training data, and use explainable AI with human-in-the-loop for high-impact decisions. Maintain audit trails, model docs, and at



Next in FinAI

AI in finance is not a magic wand—it's a disciplined workflow: define the problem, assemble quality data, select a provable model, and rigorously evaluate impact. Across fraud, credit, trading, robo-advisors, and service, measurable gains in speed, accuracy, and cost are achievable when governance and transparency anchor development. The practical design loop you practiced today can scale from a 1



Thanks!

Do you have any questions?

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